



Endometriosis: A Review of Clinical Diagnosis, Treatment, and Pathogenesis

Saurabh Chauhan¹, Akash More², Vaishnavi Chauhan¹, Aditya Kathane¹

Received 08/18/2022

Review began 08/22/2022

Review ended 08/31/2022

Published 09/06/2022

© Copyright 2022

Chauhan et al. This is an open access article distributed under the terms of the Creative Commons Attribution License CC-BY 4.0., which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited.

DOI: 10.7759/cureus.28864

1. Clinical Embryology, School of Allied Health Science, Datta Meghe Institute of Medical Sciences (DU), Wardha, IND
2. Anatomy, Wardha Test Tube Baby Center, Datta Meghe Institute of Medical Sciences (DU), Wardha, IND

Corresponding author: Akash More, aakashmore87@gmail.com

Abstract

Endometriosis is a condition that affects women of reproductive age, and it is distinguished by the development of endometrial-like tissue outside the uterine cavity. It is frequently accompanied by persistent pelvic discomfort and infertility. This investigation looks into recent findings on clinical manifestation to help doctors and improve women's health. PubMed and Google Scholar were used to review on clinical diagnosis of endometriosis. The search strategy contained the terms "endometriosis" and "clinical diagnosis." All research articles published between 1960 and 2021 were included in the search. The findings were then categorized to summarize the evidence. There was a total of 29 instances of endometriosis discovered. The patients' ages varied from 20 to 45 years old, with a median of 28.8 years and a mean of 29.4±7.7 years. Dysmenorrhea is a common disorder among adolescent girls experiencing various physical and emotional symptoms which have a detrimental influence on their quality of life. In this study, scar endometriosis was shown to be the more common variety of endometriosis, with 50% of cases predominantly developing at the lower segment cesarean section (LSCS) scar site. As a result, women with endometriosis are more likely to have miscarriages, and the quality of their embryos declines as a result.

Categories: Obstetrics/Gynecology

Keywords: infertility, endometrial, uterus, scar, endometriosis

Introduction And Background

Endometriosis is the most perplexing gynecological condition [1-3]. Endometriosis influences 10%-15% of all reproductive-age females and 70% of women with persistent pelvic pain [4]. The ovaries and pelvic peritoneum are the most common sites for developing endometriotic lesions. Endometriotic lesions can also develop in other places including the fallopian tube, abdominal wall, bowels, cervix, bladder, and vagina [5,6]. The pathophysiology of endometriosis and pain is poorly known, with most gynecologists believing that inflammation is a crucial source of irritation in endometriosis [7]. After successful surgery, some women seem to be "cured" of the disorder, but the majority will have a recurrence of symptoms. Endometriosis costs roughly 1 to 2 lakhs per woman in India, according to a survey [8]. The uterosacral ligaments, the upper third part of the posterior vaginal wall, the rectovaginal space, the intestine, and the urinary system are frequently involved in the latter [9]. Researchers have previously looked at apoptosis, cell cycle changes, and granulosa cell oxidative stress as indicators of oocyte quality as a source of endometriosis subfertility [10-12].

Endometriosis is a female reproductive system disorder where the endometrium-like tissue develops outside of the uterus; it usually affects the ovaries and peritoneum, causing premenstrual discomfort and dysmenorrhea [13,14]. The most widely recognized explanation of endometriosis is that endometrial tissue is implanted in the peritoneal cavity by retrograde menstruation. The first theory describing the origin of endometriosis is the retrograde menstruation theory. According to this idea, endometriosis develops when sloughed endometrial cells and debris after menstruation travel retrogradely down the fallopian tubes and enter the pelvic cavity. 76%-90% of women having patent fallopian tubes experience retrograde menstruation, albeit not all of these women have endometriosis [15]. Endometrial cell resorption into the abdominal wall during menstrual flow is a common occurrence in 90% of menstrual females with patent fallopian tubes, even though it is only seen in those with hormonal or immunological issues [16,17]. The peritoneal fluid of women with endometriosis has higher amounts of macrophages, T-lymphocytes, and B-lymphocytes, which are more susceptible to apoptosis [18,19].

Review

Clinical diagnosis of endometriosis

Endometriosis lesions are most frequently encountered in the following regions like fallopian tubes, uterus outer surface, ovaries, and the ligaments which surround the uterus. Lesions from endometriosis can range in size and frequently take the form of nodules or cysts. The majority of them are blue, black, and brown in color. They can, however, sometimes occasionally be white, red, or transparent [20,21]. Infertile women with minor or mild endometriosis have had the condition detected more frequently during in vitro fertilization

How to cite this article

Chauhan S, More A, Chauhan V, et al. (September 06, 2022) Endometriosis: A Review of Clinical Diagnosis, Treatment, and Pathogenesis. Cureus 14(9): e28864. DOI 10.7759/cureus.28864

(IVF) and embryo transfer (ET) cycles than those with moderate or severe endometriosis [1,22-24]. With a diagnostic age of 28 years, endometriosis predominantly affects women who are of reproductive age, which may be understood by the estrogenic milieu, which is strongly implicated in its genesis [25]. Extrauterine endometrial cells on laparoscopy are generally enough for histologic confirmation [26]. Cystic endometriomas might be reliably recognized utilizing transvaginal ultrasonography, while more modest endometrial implants are not detected [27,28].

Infertility and severe pelvic discomfort are the two major clinical risk factors for endometriosis [29-32]. The etiology of infertility in women with minor or mild endometriosis is less understood. However, it might be linked to a greater prevalence of malformed oocytes, faulty embryos, or unsuccessful implantation [33]. According to several studies, tall women with low Body Mass Index (BMI) appear at a greater risk of developing endometriosis because they have short menstrual cycles, are more likely to have cervical canalization problems, and decreased germ cell endowment [34]. Endometriosis has also been found to be less common among women who have been pregnant [29]. This might be owing to pregnancy's protective impact, or it could be due to endometriosis patients' lower fertility, as endometriosis risk is inversely associated with the number of term pregnancies [29,35,36].

Endometriosis was found in 19% of women with persistent pelvic discomfort at the point of surgery [37,38]. Dysmenorrhea, intermenstrual discomfort, and dyspareunia are common symptoms of pelvic pain. The most prevalent symptom is dysmenorrhea, which, while not predictive, is highly indicative of endometriosis in its severe form [39]. Dysmenorrhea is frequently progressive, with discomfort beginning before menstrual flow and lasting for the whole of the menstrual cycle and possibly for many days after that [40,41]. Endometriosis of the rectovaginal septum and cul-de-sac are common causes of dyspareunia [42]. Dysmenorrhea and dyspareunia are more likely to be caused by endometriosis if they appear after years of pain-free menses and coitus [43,44]. The overproduction of uterine prostaglandins aids pathogenesis. Urinary tract endometriosis can induce hematuria, urgency, dysuria, and frequency [45,46].

Dysmenorrhea

During menstruation, dysmenorrhea is a medical disorder characterized by discomfort around the pubic bone and lower abdomen [47,48]. Dysmenorrhea is related to a limitation of activity and leaves from school or work, precisely when it is severe [49]. Mental preparedness and suitable lifestyle changes, such as frequent physical activity, can help control dysmenorrhea [50]. Dysmenorrhea can be primary or secondary. Primary dysmenorrhea is a kind of dysmenorrhea that occurs in teenagers shortly after menarche and is caused by a lack of underlying macroscopic pelvic disease [51]. Early menarche, lengthier menstrual cycles, high menstrual flow, and a family history of dysmenorrhea are all risk factors [52,53].

Dyspareunia

Dyspareunia, or pain during penetration, occurs during sexual activity. Deep and superficial dyspareunia are the two most common kinds of dyspareunia [54]. Deep dyspareunia refers to the expansion of discomfort into the deeper regions of the vagina or lower pelvis, whereas superficial dyspareunia is restricted to the vulva or vaginal entrance. Profound penetration is typically correlated with deep dyspareunia [55]. Deep dyspareunia in endometriosis has a complicated etiology that might involve direct endometriotic lesions, concomitant diagnosis, and other factors [56]. Deep dyspareunia affects 50% of all female patients, a disorder that affects 10% of reproductive-age females and is linked to the poor sexual quality of life. Other forms of pelvic pain, psychiatric comorbidities, and concomitant pain diagnoses can occur in women with endometriosis [55,57].

Pathogenesis of endometriosis

Endometriosis pathogenesis suggests that the disease's etiology is complicated and multifaceted, involving genetic, hormonal, immunological, and environmental factors [58]. The pathogenesis of superficial endometriosis and the genetic factors that prevent ectopic lesions from being removed and allow for peritoneal remodeling may both be triggered by retrograde menstruation. These are critical for endometriosis lesion growth [59]. Endometriosis is spread by a change in the peritoneal fluid composition due to biological, hormone, and environmental factors [60,61]. Deep nodule tumors that are not generally eliminated during menstruation, might be quickly produced in a baboon model by transplanting basal endometrial tissue [62,63]. Endometriosis progresses due to genetic changes that damage the cell, since females with endometriosis have an abnormal endometrial cellular response, favoring extrauterine adhesion and proliferation [64]. Endometriosis has a genetic inheritance pattern that is believed to include numerous loci, as well as some chromosomal areas that have been linked to the endometriosis phenotype [65]. Genetic factors, both inherited and acquired, might predispose females to the adherence of abnormal endometrial tissues to the peritoneal epithelium as well as the resistance of immune clearing of these lesions [66]. Endometrial fragments that are still viable are pushed into the fallopian tubes by a pressure gradient caused by dyssynergic uterine contractions [67,68]. They can implant, develop, and infiltrate pelvic bones when they enter the peritoneal cavity [69,70]. Hyperalgesia is a symptom of nerve pain, which is generally caused by nerve damage or inflammatory stimulation. Endometriotic stroma cells typically invade sensory nerve fibers in deep endometriosis [71,72].

Stage	Description
I	Isolated implants characterize this minimal disease. There is no adhesions present.
II	The peritoneum and ovaries are covered in superficial implants, which are a mild form of the disease. There were no significant adhesions seen.
III	Multiple implants, both superficial and highly intrusive, make up the modern disease. Adhesions on the ovaries and fallopian tubes are possible.
IV	Severe sickness is defined by several deep and superficial implants, massive ovarian endometriomas, and other symptoms. Typically, dense adhesions are present.

TABLE 1: Stages of endometriosis

Methods

We focused on pelvic endometriosis in this study since it is the most prevalent kind of endometriosis, and it usually affects women throughout their reproductive years. PubMed and Google Scholar were used to conduct a review of the clinical diagnosis of endometriosis. In the titles or abstracts of papers, the search strategy contained the terms “endometriosis” and “clinical diagnosis.” The data from each clinical diagnosis document were extracted and categorized.

Result

In this study, there were 29 cases of endometriosis in total. The age range of the patients was 20 to 45 years, with a mean age of 29.4 ± 7.7 and a median age of 28.8 years. The scar endometriosis (16 cases) involving the anterior abdominal wall at the lower segment cesarean section (LSCS) scar site, ovarian endometriosis (10 cases), urinary bladder endometriosis (two cases), and bowel endometriosis (one case) involving the sigmoid colon. According to the data, ovarian endometriosis and scar endometriosis are the two types of endometriosis that are most common in women aged 20 to 35. Scar endometriosis has been the most frequent kind of endometriosis in this research, accounting for 50% of all cases. In 14 of the 16 patients, there was a mass in the LSCS scar, which was identified by preoperative Fine Needle Aspiration Cytology (FNAC) in nine instances (56.25%) and histological analysis of excised tissues in seven cases (43.75%). The second most prevalent condition was ovarian endometriosis, which accounted for 37.5% of cases, with one instance being bilateral and another showing torsion. Bowel endometriosis with only one case involving the sigmoid colon accounted for 4.3% of cases, whereas two cases with urinary bladder endometriosis accounted for 8.3%.

Treatment of endometriosis

The goal of medical therapy for endometriosis has been to change the menstrual period hormonally to induce a pseudopregnancy, pseudo-menopause, or persistent anovulatory state [73]. The recommended daily dose of danazol for the treatment of endometriosis is 600-800 mg; even so, the dose has significant steroid side effects, including increased hair growth, changes in mood, an irreversible deepening of the voice, bad impacts on serum lipids, and, in rare cases, irreversible and life-threatening liver injury [74]. Endometriosis is typically treated with progesterone-based medications. They induce endometrial cells to decidualize, resulting in atrophy. Abnormal uterine bleeding, vomiting, breast discomfort, and depression are all possible side effects [75,76]. Hypoestrogenism's adverse effects include vaginal heavy bleeding, vaginal dryness, reduced libido, breast discomfort, sleeplessness, and depression [77]. Endometriosis has been treated with a variety of hormones and medicines. Hormone therapy often prevents ovulation by preventing the ovaries from releasing hormones, including estrogen. This could aid in reducing the rate of local development and activity of the endometrium and endometrial lesions. Because of their negative side effects, several hormones, such as methyltestosterone and estrogen, have been phased out. Other medications have been inadequately investigated, including clomiphene, tamoxifen, and the anti-progestational drug mifepristone [78]. Laparoscopic surgery is used to surgically diagnose endometriosis. A laparoscope, a narrow viewing tube, is inserted into the belly through a tiny incision during laparoscopy. For a second entrance for the surgical tools, a second incision may be created in the lower abdomen. Your doctor can view the outside of the ovaries, uterus, fallopian tubes, and other organs up close using a laparoscope. Surgical tools for removing scar tissue or obtaining tissue samples can also be attached to the laparoscope [79]. For ovarian cysts greater than 4 to 5 cm in diameter, laparoscopic endometrioma removal is suggested [80]. Endometriosis has been treated with a range of devices, from monopolar cautery and scissors to a wide variety of lasers and ultrasonic scalpels [81]. Endometriosis pain is treated surgically by interrupting the brain circuits that carry pain signals [82]. Various considerations should be taken, including the best method of gaining access to the pelvis and abdomen, which can be accomplished by laparoscopy or laparotomy. Laparoscopy is less expensive and takes less time to recuperate. Magnetic resonance imaging (MRI) scan not only shows morphologic defects in the bladder but can also identify other common areas, such as the uterosacral joints, where ultrasonography is less accurate [83].

Discussion

Endometriosis affects people and cultures all over the world; however, there are significant delays in diagnosis. More illness awareness in inpatient healthcare should result in earlier diagnosis, less suffering, and increased productivity [84,85]. Gravidity, endometriosis in the family, dyspareunia, tiredness, dysmenorrhea, pelvic surgery, diarrhea, pelvic discomfort, and premenstrual spotting are all risk factors for endometriosis [86]. Several explanations for the connection between endometriosis and infertility have been claimed, but despite substantial investigation, no conclusion has been achieved. These mechanisms include distorted pelvic anatomy, altered peritoneal function, endocrine, and ovulatory abnormalities, and altered hormonal and cell-mediated functions in the endometrium [87,88]. Increased obstetric surgery results in the direct mechanical implantation of endometrial cells, which is one of the major risk factors for scar endometriosis [89].

When a patient experiences dysmenorrhea after experiencing pain-free menstrual cycles for years, endometriosis should be suspected [90]. Endometriomas having a maximal diameter of 3-4 cm are best treated through transvaginal cystectomy. Less than 1% of women develop ovarian endometriosis malignancy, which most frequently manifests as endometrioid/clear cell carcinoma [91]. Between 1% and 5% of women with endometriosis have urinary tract involvement, with the urinary bladder being the most commonly affected, followed by the ureter, urethra, and kidney [92,93]. In 3%-37% of instances, endometriosis was known to have an impact on the digestive system, with the recto-sigmoid colon as the most often affected area [94]. Although comparable findings have been documented in cancer patients, the link between diagnostic delay and health care funding for endometriosis is unique [95-97]. The majority of endometriosis patients have normal pelvic findings; hence a laparoscopy is required to make a conclusive diagnosis. It is likely that ultimately a combination of biochemical indicators and clinical evaluation will lessen the requirement for surgical confirmation, even if no single laboratory test has demonstrated proven clinical usefulness [98,99]. Endometriosis is more frequent among Caucasian, middle-aged, upper-class women who are ambitious. This could happen because these women have more access to medical treatment and diagnostic procedures like laparoscopy [100]. This might be because these women have easier access to medical treatment and diagnostic testing like laparoscopy [101,102]. As endometriosis frequently exhibits symptoms that resemble those of other disorders and cause a diagnostic delay, early suspicion of the condition is crucial for prompt diagnosis. In addition to a patient's medical history, secondary centers with access to further investigations may need to be referred from the main health care level [103,104].

Ovarian Endometriosis

Ovarian endometriomas account for 35% of all benign ovarian cysts and are seen in 17%-44% of female patients [105]. Endometrial cell growth may be induced by ovarian follicular fluid [106]. Left ovary endometriomas are more prevalent than right ovary endometriomas [107]. The amount of normal ovarian in the swollen ovarian cortex was decreased in endometrioma patients' ovarian cortical tissue, a result not seen in other benign cysts to a similar extent [108]. The ovary with endometrial cysts exhibits lower reactivity to exogenous gonadotropin activation, with more follicular atresia, and a higher rate of follicular atrophy [109]. The most prevalent histologic forms of ovarian cancer resulting from ovarian endometriosis are endometrioid adenocarcinoma and clear cell carcinoma [110,111].

Scar Endometriosis

Scar endometriosis occurs when endometrial tissue is implanted directly in scars during surgery [112]. Other surgical disorders such as hematoma, hernia, scar tissue, neuroma, abscess, granuloma, or even metastatic cancer can easily be coupled with scar endometriosis [113,114]. The presence of morphological and physiological stroma and endometrial gland beyond the uterine cavity is a persistent gynecologic condition [115]. The kidneys, pleura, bladder, lungs, colon, lymph nodes, omentum, and abdominal wall are all common locations for extra pelvic endometriosis [116].

Urinary Bladder Endometriosis

Extragenital endometriosis most commonly affects the gut and urinary system [117]. The urinary bladder is a rare location of endometrial localization, with just around 1% of people that are suffering from the illness having lesions affecting the urine system [118]. To avoid kidney function loss, urinary tract endometriosis must be diagnosed and treated early. For women who complain of bladder pain but have a negative urine test, the doctor may not examine if the discomfort is cyclic or whether bladder endometriosis is a possibility. Endometriosis in the urinary system is uncommon, occurring in about 1%-6% of females with initially diagnosed endometriosis [119].

Conclusions

Long-term conditions like endometriosis with symptoms like dyspareunia, dysmenorrhea, dysuria, dyschezia, and chronic pelvic pain (CPP) vary based on the organ affected, but not the illness itself. For early identification and management of bladder, ovarian, and intestine endometriosis, a higher index of concern is essential to improve the quality of life and decrease infertility. Dysmenorrhea is a fairly prevalent condition in adolescent girls, as they suffer from a variety of mental and physical symptoms that negatively

impact their quality of life. Scar endometriosis has been the most common site of endometriosis in this research, with 50% of cases developing solely at the LSCS scar site. Minimal peritoneal endometriosis and also the most severe phases of the disease can be treated with laparoscopic surgery. Therefore, women with endometriosis are at a higher risk of miscarriage, and their embryo quality suffers.

Additional Information

Disclosures

Conflicts of interest: In compliance with the ICMJE uniform disclosure form, all authors declare the following: **Payment/services info:** All authors have declared that no financial support was received from any organization for the submitted work. **Financial relationships:** All authors have declared that they have no financial relationships at present or within the previous three years with any organizations that might have an interest in the submitted work. **Other relationships:** All authors have declared that there are no other relationships or activities that could appear to have influenced the submitted work.

References

- Xu B, Guo N, Zhang XM, Shi W, Tong XH, Iqbal F, Liu YS: Oocyte quality is decreased in women with minimal or mild endometriosis. *Sci Rep*. 2015, 5:10779. [10.1038/srep10779](https://doi.org/10.1038/srep10779)
- Sourial S, Tempest N, Hapangama DK: Theories on the pathogenesis of endometriosis. *Int J Reprod Med*. 2014, 2014:179515. [10.1155/2014/179515](https://doi.org/10.1155/2014/179515)
- Collins JA, Burrows EA, Willan AR: The prognosis for live birth among untreated infertile couples. *Fertil Steril*. 1995, 64:22-8. [10.1016/s0015-0282\(16\)57650-x](https://doi.org/10.1016/s0015-0282(16)57650-x)
- Parasar P, Ozcan P, Terry KL: Endometriosis: epidemiology, diagnosis and clinical management. *Curr Obstet Gynecol Rep*. 2017, 6:34-41. [10.1007/s13669-017-0187-1](https://doi.org/10.1007/s13669-017-0187-1)
- Farland L v, Shah DK, Kvaskoff M, et al.: Epidemiological and clinical risk factors for endometriosis. D'Hooghe T (ed): Springer, Cham; 2017. [10.1007/978-3-319-59856-7_6](https://doi.org/10.1007/978-3-319-59856-7_6)
- Wang G, Tokushige N, Markham R, Fraser IS: Rich innervation of deep infiltrating endometriosis. *Hum Reprod*. 2009, 24:827-34. [10.1093/humrep/den464](https://doi.org/10.1093/humrep/den464)
- Vercellini P, Somigliana E, Viganò P, Abbiati A, Daguati R, Crosignani PG: Endometriosis: current and future medical therapies. *Best Pract Res Clin Obstet Gynaecol*. 2008, 22:275-306. [10.1016/j.bpobgyn.2007.10.001](https://doi.org/10.1016/j.bpobgyn.2007.10.001)
- Moradi M, Parker M, Sneddon A, Lopez V, Ellwood D: Impact of endometriosis on women's lives: a qualitative study. *BMC Womens Health*. 2014, 14:123. [10.1186/1472-6874-14-123](https://doi.org/10.1186/1472-6874-14-123)
- Cirstoiu M, Bodean O, Secara D, Munteanu O, Cirstoiu C: Case study of a rare form of endometriosis. *J Med Life*. 2013, 6:68-71.
- Sanchez AM, Vanni VS, Bartiromo L, et al.: Is the oocyte quality affected by endometriosis? A review of the literature. *J Ovarian Res*. 2017, 10:43. [10.1186/s13048-017-0341-4](https://doi.org/10.1186/s13048-017-0341-4)
- Ohta N, Saito H, Kuzumaki T, et al.: Expression of CD44 in human cumulus and mural granulosa cells of individual patients in in-vitro fertilization programmes. *Mol Hum Reprod*. 1999, 5:22-8. [10.1093/molehr/5.1.22](https://doi.org/10.1093/molehr/5.1.22)
- Nakahara K, Saito H, Saito T, et al.: Ovarian fecundity in patients with endometriosis can be estimated by the incidence of apoptotic bodies. *Fertil Steril*. 1998, 69:931-5. [10.1016/S0015-0282\(98\)00038-7](https://doi.org/10.1016/S0015-0282(98)00038-7)
- Eisenberg VH, Weil C, Chodick G, Shalev V: Epidemiology of endometriosis: a large population-based database study from a healthcare provider with 2 million members. *BJOG*. 2018, 125:55-62. [10.1111/1471-0528.14711](https://doi.org/10.1111/1471-0528.14711)
- Melis I, Litta P, Nappi L, et al.: Sexual function in women with deep endometriosis: correlation with quality of life, intensity of pain, depression, anxiety, and body image. *Int J Sexual Health*. 2015, 27:175-85. [10.1080/19317611.2014.952394](https://doi.org/10.1080/19317611.2014.952394)
- Brosens I, Gargett CE, Guo SW, Puttemans P, Gordts S, Brosens JJ, Benagiano G: Origins and progression of adolescent endometriosis. *Reprod Sci*. 2016, 23:1282-8. [10.1177/1933719116637919](https://doi.org/10.1177/1933719116637919)
- Anaf V, Simon P, El Nakadi I, et al.: Relationship between endometriotic foci and nerves in rectovaginal endometriotic nodules. *Hum Reprod*. 2000, 15:1744-50. [10.1093/humrep/15.8.1744](https://doi.org/10.1093/humrep/15.8.1744)
- Ballester M, Gonin J, Rodenas A, Bernaudin JF, Rouzier R, Coutant C, Daraï E: Eutopic endometrium and peritoneal, ovarian and colorectal endometriotic tissues express a different profile of nectin-1, -3, -4 and nectin-like molecule 2. *Hum Reprod*. 2012, 27:3179-86. [10.1093/humrep/des304](https://doi.org/10.1093/humrep/des304)
- Prescott J, Farland LV, Tobias DK, et al.: A prospective cohort study of endometriosis and subsequent risk of infertility. *Hum Reprod*. 2016, 31:1475-82. [10.1093/humrep/dew085](https://doi.org/10.1093/humrep/dew085)
- Shim JY, Laufer MR: Adolescent endometriosis: an update. *J Pediatr Adolesc Gynecol*. 2020, 33:112-9. [10.1016/j.jpag.2019.11.011](https://doi.org/10.1016/j.jpag.2019.11.011)
- Ozkan S, Murk W, Arici A: Endometriosis and infertility: epidemiology and evidence-based treatments. *Ann N Y Acad Sci*. 2008, 1127:92-100. [10.1196/annals.1434.007](https://doi.org/10.1196/annals.1434.007)
- Damewood MD: The role of the new reproductive technologies including IVF and GIFT in endometriosis. *Obstet Gynecol Clin North Am*. 1989, 16:179-91. [10.1016/s0889-8545\(21\)00146-7](https://doi.org/10.1016/s0889-8545(21)00146-7)
- Harb HM, Gallos ID, Chu J, Harb M, Coomarasamy A: The effect of endometriosis on in vitro fertilisation outcome: a systematic review and meta-analysis. *BJOG*. 2013, 120:1308-20. [10.1111/1471-0528.12366](https://doi.org/10.1111/1471-0528.12366)
- Suzuki T, Izumi S, Matsubayashi H, Awaji H, Yoshikata K, Makino T: Impact of ovarian endometrioma on oocytes and pregnancy outcome in in vitro fertilization. *Fertil Steril*. 2005, 83:908-13. [10.1016/j.fertnstert.2004.11.028](https://doi.org/10.1016/j.fertnstert.2004.11.028)
- Barnhart K, Dunsmoor-Su R, Coutifaris C: Effect of endometriosis on in vitro fertilization. *Fertil Steril*. 2002, 77:1148-55. [10.1016/S0015-0282\(02\)03112-6](https://doi.org/10.1016/S0015-0282(02)03112-6)
- Pritts EA, Taylor RN: An evidence-based evaluation of endometriosis-associated infertility. *Endocrinol Metab Clin North Am*. 2003, 32:653-67. [10.1016/S0889-8529\(03\)00045-8](https://doi.org/10.1016/S0889-8529(03)00045-8)

26. Schrager S, Falleroni J, Edgoose J: Evaluation and treatment of endometriosis . *Am Fam Physician*. 2013, 87:107-13.
27. Luna Russo MA, Chalif JN, Falcone T: Clinical management of endometriosis. *Minerva Ginecol*. 2020, 72:106-18. [10.23736/S0026-4784.20.04544-X](#)
28. Alcázar JL, Laparte C, Jurado M, et al.: The role of transvaginal ultrasonography combined with color velocity imaging and pulsed Doppler in the diagnosis of endometrioma. *Fertil Steril*. 1997, 67:487-91. [10.1016/S0015-0282\(97\)80074-X](#)
29. Kiesel L, Sourouni M: Diagnosis of endometriosis in the 21st century . *Climacteric*. 2019, 22:296-302. [10.1080/13697137.2019.1578743](#)
30. Strathy JH, Molgaard CA, Coulam CB, et al.: Endometriosis and infertility: a laparoscopic study of endometriosis among fertile and infertile women. *Fertil Steril*. 1985, 44:667-72. [10.1016/s0015-0282\(16\)48788-1](#)
31. Mahmood TA, Templeton A: Prevalence and genesis of endometriosis. *Hum Reprod*. 1991, 6:544-9. [10.1093/oxfordjournals.humrep.a137377](#)
32. el-Yahia AW: Laparoscopic evaluation of apparently normal infertile women . *Aust N Z J Obstet Gynaecol*. 1994, 34:440-2. [10.1111/j.1479-828x.1994.tb01266.x](#)
33. Nakamura T: Clinical aspects of adolescent endometriosis . *Endocrines*. 2021, 2:301-10. [10.3390/endocrines2030028](#)
34. Signorello LB, Harlow BL, Cramer DW, et al.: Epidemiologic determinants of endometriosis: a hospital-based case- control study. *Ann Epidemiol*. 1997, 7:267-74. [10.1016/S1047-2797\(97\)00017-3](#)
35. Sangi-Haghpeykar H, Poindexter AN: Epidemiology of endometriosis among parous women . *Obstet Gynecol*. 1995, 85:983-92. [10.1016/0029-7844\(95\)00074-2](#)
36. Parazzini F, Ardovino I, Struzziero E, et al.: Risk factors for pelvic endometriosis in women with pelvic pain or infertility. *Eur J Obstet Gynecol Reprod Biol*. 1999, 83:195-9. [10.1016/S0301-2115\(98\)00332-7](#)
37. Kresch AJ, Seifer DB, Sachs LB, et al.: Laparoscopy in 100 women with chronic pelvic pain . *Obstet Gynecol*. 1984, 64:672-4. [10.1016/s0196-0644\(85\)80300-0](#)
38. Galle PC: Clinical presentation and diagnosis of endometriosis. *Obstet Gynecol Clin North Am*. 1989, 16:29-42. [10.1016/s0889-8545\(21\)00136-4](#)
39. Mahmood TA, Templeton AA, Thomson L, Fraser C: Menstrual symptoms in women with pelvic endometriosis. *Br J Obstet Gynaecol*. 1991, 98:558-63. [10.1111/j.1471-0528.1991.tb10370.x](#)
40. Davis GD, Thillet E, Lindemann J: Clinical characteristics of adolescent endometriosis . *J Adolesc Health*. 1993, 14:362-8. [10.1016/S1054-139X\(08\)80008-0](#)
41. Parazzini F, Cipriani S, Moroni S, et al.: Relationship between stage, site and morphological characteristics of pelvic endometriosis and pain. *Hum Reprod*. 2001, 16:2668-71. [10.1093/humrep/16.12.2668](#)
42. Vercellini P, Trespidi L, de Giorgi O, et al.: Endometriosis and pelvic pain: Relation to disease stage and localization. *Fertil Steril*. 1996, 65:299-304. [10.1016/s0015-0282\(16\)58089-3](#)
43. Fedele L, Bianchi S, Bociolone L, Di Nola G, Parazzini F: Pain symptoms associated with endometriosis . *Obstet Gynecol*. 1992, 79:767-9.
44. Ranney B: Endometriosis. II. Emergency operations due to hemoperitoneum . *Obstet Gynecol*. 1970, 36:437-42.
45. Brenner C, Wohlgemuth S: Scar endometriosis. *Surg Gynecol Obstet*. 1990, 170:538-40. [10.37549/ar2541](#)
46. Chingwundoh FI, Ryan P, Luesley T, Chan SY: Renal and diaphragmatic endometriosis de novo associated with hormone replacement therapy. *J Urol*. 1995, 153:380-1. [10.1097/00005392-199502000-00025](#)
47. Olubunmi OP, Yinka OS, Oladele OJ, et al.: A case study of the prevalence of dysmenorrhea and its effects among females of different age groups. *J Exp Integr Med*. 2016, 6:125. [10.5455/jeim.180916.or.161](#)
48. Kim SS: Studies on pre-modern medical history in Korea, 2010-2019: Increased study areas and diversified approaches. *Uisahak*. 2020, 29:371-423. [10.13081/kjmh.2020.29.371](#)
49. Weissman AM, Hartz AJ, Hansen MD, Johnson SR: The natural history of primary dysmenorrhea: a longitudinal study. *BJOG*. 2004, 111:345-52. [10.1111/j.1471-0528.2004.00090.x](#)
50. Agarwal AK, Agarwal A: A study of dysmenorrhea during menstruation in adolescent girls . *Indian J Community Med*. 2010, 35:159-64. [10.4103/0970-0218.62586](#)
51. Gupta S, Kaur S, Sadiq S, et al.: Primary dysmenorrhea: evaluation and treatment pattern among female medical students. *Int J Basic Clinic Pharmacol*. 2018, 7: [10.18203/2319-2003.ijbcp20183883](#)
52. Chiu MH, Hsieh HF, Yang YH, Chen HM, Hsu SC, Wang HH: Influencing factors of dysmenorrhea among hospital nurses: a questionnaire survey in Taiwan. *BMJ Open*. 2017, 7:e017615. [10.1136/bmjopen-2017-017615](#)
53. Dawood MY: Dysmenorrhea and prostaglandins: pharmacological and therapeutic considerations. *Drugs*. 1981, 22:42-56. [10.2165/00003495-198122010-00003](#)
54. Shum LK, Bedaiwy MA, Allaire C, et al.: Deep dyspareunia and sexual quality of life in women with endometriosis. *Sex Med*. 2018, 6:224-33. [10.1016/j.esxm.2018.04.006](#)
55. Yong PJ, Sadownik L, Brotto LA: Concurrent deep-superficial dyspareunia: prevalence, associations, and outcomes in a multidisciplinary vulvodynia program. *J Sex Med*. 2015, 12:219-27. [10.1111/jsm.12729](#)
56. Yong PJ: Deep dyspareunia in endometriosis: a proposed framework based on pain mechanisms and genito-pelvic pain penetration disorder. *Sex Med Rev*. 2017, 5:495-507. [10.1016/j.sxm.2017.06.005](#)
57. Brotto LA, Yong P, Smith KB, Sadownik LA: Impact of a multidisciplinary vulvodynia program on sexual functioning and dyspareunia. *J Sex Med*. 2015, 12:238-47. [10.1111/jsm.12718](#)
58. Nap AW: Theories on the pathogenesis of endometriosis . *Endometriosis: Science and Practice*. Giudice LC (ed): Blackwell Publishing Ltd, England; 2012. [10.1002/9781444398519.ch5](#)
59. Nasu K, Yuge A, Tsuno A, Nishida M, Narahara H: Involvement of resistance to apoptosis in the pathogenesis of endometriosis. *Histol Histopathol*. 2009, 24:1181-92. [10.14670/HH-24.1181](#)
60. Gilabert-Estelles J, Ramon LA, España F, Gilabert J, Castello R, Estelles A: Expression of the fibrinolytic components in endometriosis. *Pathophysiol Haemost Thromb*. 2006, 35:136-40. [10.1159/000093556](#)
61. Szczepańska M, Koźlik J, Skrzypczak J, et al.: Oxidative stress may be a piece in the endometriosis puzzle . *Fertil Steril*. 2003, 79:1288-93. [10.1016/S0015-0282\(03\)00266-8](#)

62. Dehoux JP, Defrère S, Squifflet J, et al.: Is the baboon model appropriate for endometriosis studies? . *Fertil Steril*. 2011, 96:728-33.e3. [10.1016/j.fertnstert.2011.06.037](https://doi.org/10.1016/j.fertnstert.2011.06.037)
63. Donnez O, Van Langendonck A, Defrère S, et al.: Induction of endometriotic nodules in an experimental baboon model mimicking human deep nodular lesions. *Fertil Steril*. 2013, 99:783-9.e3. [10.1016/j.fertnstert.2012.10.032](https://doi.org/10.1016/j.fertnstert.2012.10.032)
64. Koninckx PR, Barlow D, Kennedy S: Implantation versus infiltration: the Sampson versus the endometriotic disease theory. *Gynecol Obstet Invest*. 1999, 47 Suppl 1:3-9; discussion 9-10. [10.1159/000052853](https://doi.org/10.1159/000052853)
65. Albertsen HM, Chettier R, Farrington P, Ward K: Genome-wide association study link novel loci to endometriosis. *PLoS One*. 2013, 8:e58257. [10.1371/journal.pone.0058257](https://doi.org/10.1371/journal.pone.0058257)
66. Burney RO, Giudice LC: Pathogenesis and pathophysiology of endometriosis . *Fertil Steril*. 2012, 98:511-9. [10.1016/j.fertnstert.2012.06.029](https://doi.org/10.1016/j.fertnstert.2012.06.029)
67. Sznurkowski JJ, Emerich J: Endometriomas are more frequent on the left side . *Acta Obstet Gynecol Scand*. 2008, 87:104-6. [10.1080/00016340701671929](https://doi.org/10.1080/00016340701671929)
68. Parazzini F, Ferraroni M, Fedele L, Boccione L, Rubessa S, Riccardi A: Pelvic endometriosis: reproductive and menstrual risk factors at different stages in Lombardy, northern Italy. *J Epidemiol Community Health*. 1995, 49:61-4. [10.1136/jech.49.1.61](https://doi.org/10.1136/jech.49.1.61)
69. Moen MH, Magnus P: The familial risk of endometriosis . *Acta Obstet Gynecol Scand*. 1993, 72:560-4. [10.3109/00016349309058164](https://doi.org/10.3109/00016349309058164)
70. Frisch RE, Wyshak G, Albert LS, Sober AJ: Dysplastic nevi, cutaneous melanoma, and gynecologic disorders. *Int J Dermatol*. 1992, 31:331-5. [10.1111/j.1365-4362.1992.tb03948.x](https://doi.org/10.1111/j.1365-4362.1992.tb03948.x)
71. Cramer DW, Wilson E, Stillman RJ, et al.: The relation of endometriosis to menstrual characteristics, smoking, and exercise. *JAMA*. 1986, 255:1904-8. [10.1001/jama.1986.03370140102032](https://doi.org/10.1001/jama.1986.03370140102032)
72. Guntupalli SR, Doo DW, Guy M, et al.: Preparedness of obstetrics and gynecology residents for fellowship training. *Obstet Gynecol*. 2015, 126:559-68. [10.1097/AOG.0000000000000999](https://doi.org/10.1097/AOG.0000000000000999)
73. Melega C, Balducci M, Bulletti C, Galassi A, Jasonni VM, Flamigni C: Tissue factors influencing growth and maintenance of endometriosis. *Ann N Y Acad Sci*. 1991, 622:256-65. [10.1111/j.1749-6632.1991.tb37869.x](https://doi.org/10.1111/j.1749-6632.1991.tb37869.x)
74. Buttram VC, Belue JB, Reiter R: Interim report of a study of danazol for the treatment of endometriosis . *Fertil Steril*. 1982, 37:478-83. [10.1016/S0015-0282\(16\)46151-0](https://doi.org/10.1016/S0015-0282(16)46151-0)
75. Razzi S, Luisi S, Calonaci F, Altomare A, Bocchi C, Petraglia F: Efficacy of vaginal danazol treatment in women with recurrent deeply infiltrating endometriosis. *Fertil Steril*. 2007, 88:789-94. [10.1016/j.fertnstert.2006.12.077](https://doi.org/10.1016/j.fertnstert.2006.12.077)
76. Berlanda N, Somigliana E, Viganò P, Vercellini P: Safety of medical treatments for endometriosis. *Expert Opin Drug Saf*. 2016, 15:21-30. [10.1517/14740358.2016.1121991](https://doi.org/10.1517/14740358.2016.1121991)
77. Martone S, Troia L, Marcolongo P, Luisi S: Role of medical treatment of endometriosis . *Minerva Obstet Gynecol*. 2021, 73:304-16. [10.23736/S2724-606X.21.04784-5](https://doi.org/10.23736/S2724-606X.21.04784-5)
78. Balasch J, Creus M, Fábregues F, Carmona F, Martínez-Román S, Manau D, Vanrell JA: Pentoxifylline versus placebo in the treatment of infertility associated with minimal or mild endometriosis: a pilot randomized clinical trial. *Hum Reprod*. 1997, 12:2046-50. [10.1093/humrep/12.9.2046](https://doi.org/10.1093/humrep/12.9.2046)
79. Busacca M, Fedele L, Bianchi S, Candiani M, Agnoli B, Raffaelli R, Vignali M: Surgical treatment of recurrent endometriosis: laparotomy versus laparoscopy. *Hum Reprod*. 1998, 13:2271-4. [10.1093/humrep/13.8.2271](https://doi.org/10.1093/humrep/13.8.2271)
80. Catalano GF, Marana R, Caruana P, et al.: Laparoscopy versus microsurgery by laparotomy for excision of ovarian cysts in patients with moderate or severe endometriosis. *J Am Assoc Gynecol Laparosc*. 1996, 3:267-70. [10.1016/S1074-3804\(96\)80011-9](https://doi.org/10.1016/S1074-3804(96)80011-9)
81. Tulandi T, al-Took S: Reproductive outcome after treatment of mild endometriosis with laparoscopic excision and electrocoagulation. *Fertil Steril*. 1998, 69:229-31. [10.1016/s0015-0282\(97\)00469-x](https://doi.org/10.1016/s0015-0282(97)00469-x)
82. Wei Y, Liang Y, Lin H, Dai Y, Yao S: Autonomic nervous system and inflammation interaction in endometriosis-associated pain. *J Neuroinflammation*. 2020, 17:80. [10.1186/s12974-020-01752-1](https://doi.org/10.1186/s12974-020-01752-1)
83. Fedele L, Bianchi S, Raffaelli R, Portuese A: Pre-operative assessment of bladder endometriosis . *Hum Reprod*. 1997, 12:2519-22. [10.1093/humrep/12.11.2519](https://doi.org/10.1093/humrep/12.11.2519)
84. Johnson NP, Hummelshoj L, Adamson GD, et al.: World Endometriosis Society consensus on the classification of endometriosis. *Hum Reprod*. 2017, 32:315-24. [10.1093/humrep/dew293](https://doi.org/10.1093/humrep/dew293)
85. Culley L, Law C, Hudson N, Mitchell H, Denny E, Raine-Fenning N: A qualitative study of the impact of endometriosis on male partners. *Hum Reprod*. 2017, 32:1667-73. [10.1093/humrep/dex221](https://doi.org/10.1093/humrep/dex221)
86. Ashrafi M, Sadatmahalleh SJ, Akhoond MR, et al.: Evaluation of risk factors associated with endometriosis in infertile women. *Int J Fertil Steril*. 2016, 10:11-21. [10.22074/2Fijfs.2016.4763](https://doi.org/10.22074/2Fijfs.2016.4763)
87. Agarwal N, Subramanian A: Endometriosis - morphology, clinical presentations and molecular pathology . *J Lab Physicians*. 2010, 2:1-9. [10.4103/0974-2727.66699](https://doi.org/10.4103/0974-2727.66699)
88. Macer ML, Taylor HS: Endometriosis and infertility: a review of the pathogenesis and treatment of endometriosis-associated infertility. *Obstet Gynecol Clin North Am*. 2012, 39:535-49. [10.1016/j.ogc.2012.10.002](https://doi.org/10.1016/j.ogc.2012.10.002)
89. Leite GK, Carvalho LF, Korkes H, Guazzelli TF, Kenj G, Viana Ade T: Scar endometrioma following obstetric surgical incisions: retrospective study on 33 cases and review of the literature. *Sao Paulo Med J*. 2009, 127:270-7. [10.1590/s1516-31802009000500005](https://doi.org/10.1590/s1516-31802009000500005)
90. Patil NJ, Kumar V, Gupta A: Scar endometriosis-a sequel of caesarean section. *J Clin Diagn Res*. 2014, 8:FD09-10. [10.7860/JCDR/2014/7554.4267](https://doi.org/10.7860/JCDR/2014/7554.4267)
91. Karaman K, Pala EE, Bayol U, Akman O, Olmez M, Unluoglu S, Ozturk S: Endometriosis of the terminal ileum: a diagnostic dilemma. *Case Rep Pathol*. 2012, 2012:742035. [10.1155/2012/742035](https://doi.org/10.1155/2012/742035)
92. Pastor JF, Diego RB, Gómez MÁC, Díez ET, Flórez AF, Olmos GB, Baños JLG: Bladder endometriosis and endocervicosis: presentation of 2 cases with endoscopic management and review of literature. *Case Rep Urol*. 2014, 2014:296908. [10.1155/2014/296908](https://doi.org/10.1155/2014/296908)
93. Maccagnano C, Pellucchi F, Rocchini L, et al.: Diagnosis and treatment of bladder endometriosis: state of the art. *Urol Int*. 2012, 89:249-58. [10.1159/000339519](https://doi.org/10.1159/000339519)
94. Badipatla KR, Vupputuri A, Niazi M, Blaise MN, Nayudu SK: Colonic endometriosis: dig deeper for diagnosis. *Gastroenterology Res*. 2017, 10:59-62. [10.14740/gr760e](https://doi.org/10.14740/gr760e)

95. Marrie RA, Horwitz R, Cutter G, Tyry T, Campagnolo D, Vollmer T: Comorbidity delays diagnosis and increases disability at diagnosis in MS. *Neurology*. 2009, 72:117-24. [10.1212/01.wnl.0000333252.78173.5f](#)
96. Arndt V, Stürmer T, Stegmaier C, Ziegler H, Dhom G, Brenner H: Patient delay and stage of diagnosis among breast cancer patients in Germany -- a population based study. *Br J Cancer*. 2002, 86:1034-40. [10.1038/sj.bjc.6600209](#)
97. Martin S, Ulrich C, Munsell M, Taylor S, Lange G, Bleyer A: Delays in cancer diagnosis in underinsured young adults and older adolescents. *Oncologist*. 2007, 12:816-24. [10.1634/theoncologist.12-7-816](#)
98. Valle RF: Endometriosis: current concepts and therapy. *Int J Gynaecol Obstet*. 2002, 78:107-19. [10.1016/S0020-7292\(02\)00132-7](#)
99. Gustilo-Ashby AM, Paraíso MF: Treatment of urinary tract endometriosis. *J Minim Invasive Gynecol*. 2006, 13:559-65. [10.1016/j.jmig.2006.07.012](#)
100. Hediger ML, Hartnett HJ, Louis GM: Association of endometriosis with body size and figure. *Fertil Steril*. 2005, 84:1366-74. [10.1016/j.fertnstert.2005.05.029](#)
101. Bonocher CM, Montenegro ML, Rosa E Silva JC, Ferriani RA, Meola J: Endometriosis and physical exercises: a systematic review. *Reprod Biol Endocrinol*. 2014, 12:4. [10.1186/1477-7827-12-4](#)
102. Kennedy S: Who gets endometriosis?. *Women's Health Med*. 2005, 2:18-9. [10.1383/wohm.2.1.18.58876](#)
103. Ballard KD, Mangubat MC: 24: Can Symptomatology Provide Clues to the Diagnosis of Endometriosis?: A National Case-control Study of 5,544 Women With Endometriosis. *J Minimally Invasive Gynecol*. 2007, 14:[10.1016/j.jmig.2007.08.025](#)
104. Abbas S, Ihle P, Köster I, Schubert I: Prevalence and incidence of diagnosed endometriosis and risk of endometriosis in patients with endometriosis-related symptoms: findings from a statutory health insurance-based cohort in Germany. *Eur J Obstet Gynecol Reprod Biol*. 2012, 160:79-83. [10.1016/j.ejogrb.2011.09.041](#)
105. Gordts S, Puttemans P, Gordts S, Brosens I: Ovarian endometrioma in the adolescent: a plea for early-stage diagnosis and full surgical treatment. *Gynecol Surg*. 2015, 12:21-30. [10.1007/s10397-014-0877-x](#)
106. Benagiano G, Bianchi P, Brosens I: Ovarian endometriomas in adolescents often represent active angiogenic disease requiring early diagnosis and careful management. *Minerva Ginecol*. 2017, 69:100-7. [10.23736/S0026-4784.16.03919-8](#)
107. Suryawanshi S, Huang X, Elishaev E, et al.: Complement pathway is frequently altered in endometriosis and endometriosis-associated ovarian cancer. *Clin Cancer Res*. 2014, 20:6163-74. [10.1158/1078-0432.CCR-14-1338](#)
108. Lee SY, Kim ML, Seong SJ, Bae JW, Cho YJ: Recurrence of ovarian endometrioma in adolescents after conservative, laparoscopic cyst enucleation. *J Pediatr Adolesc Gynecol*. 2017, 30:228-33. [10.1016/j.jpog.2015.11.001](#)
109. Kitajima M, Defrère S, Dolmans MM, Colette S, Squifflet J, Van Langendonck A, Donnez J: Endometriomas as a possible cause of reduced ovarian reserve in women with endometriosis. *Fertil Steril*. 2011, 96:685-91. [10.1016/j.fertnstert.2011.06.064](#)
110. Filippi F, Benaglia L, Paffoni A, Restelli L, Vercellini P, Somigliana E, Fedele L: Ovarian endometriomas and oocyte quality: insights from in vitro fertilization cycles. *Fertil Steril*. 2014, 101:988-93.e1. [10.1016/j.fertnstert.2014.01.008](#)
111. Gałczyński K, Jóźwik M, Lewkowicz D, Semczuk-Sikora A, Semczuk A: Ovarian endometrioma - a possible finding in adolescent girls and young women: a mini-review. *J Ovarian Res*. 2019, 12:104. [10.1186/s13048-019-0582-5](#)
112. Scully RE, Barlow JF: Mesonephroma: of ovary. Tumor of müllerian nature related to the endometrioid carcinoma. *Cancer*. 1967, 20:1405-17. [10.1002/1097-0142\(196709\)20:9%3C1405::aid-cnrcr2820200907%3E3.0.co;2-b](#)
113. Blanco RG, Parthivel VS, Shah AK, et al.: Abdominal wall endometriomas. *Am J Surg*. 2003, 185:596-8. [10.1016/S0002-9610\(03\)00072-2](#)
114. Bektaş H, Bilsel Y, Sari YS, et al.: Abdominal wall endometrioma; a 10-year experience and brief review of the literature. *J Surg Res*. 2010, 164:e77-81. [10.1016/j.jss.2010.07.043](#)
115. Chauhan SP, Berghella V, Sanderson M, Magann EF, Morrison JC: American College of Obstetricians and Gynecologists practice bulletins: an overview. *Am J Obstet Gynecol*. 2006, 194:1564-72; discussion 1072-5. [10.1016/j.ajog.2006.03.001](#)
116. Khachani I, Filali Adib A, Beza R: Cesarean scar endometriosis: an uncommon surgical complication on the rise? Case report and literature review. *Case Rep Obstet Gynecol*. 2017, 2017:8062924. [10.1155/2017/8062924](#)
117. Soyinka AS: L Mettler (ed): Manual for laparoscopic and hysteroscopic gynaecological surgery. *Gynecol Surg*. 2007, 4:65. [10.1007/s10397-006-0248-3](#)
118. Veeraswamy A, Lewis M, Mann A, Kotikela S, Hajhosseini B, Nezhat C: Extragenital endometriosis. *Clin Obstet Gynecol*. 2010, 53:449-66. [10.1097/GRF.0b013e3181e0ea6e](#)
119. Kołodziej A, Krajewski W, Dołowy Ł, et al.: Urinary tract endometriosis. *Urol J*. 2015, 12:2213-7. [10.1177/228402650900100202](#)